

Deep Learning - 5 Page Study Guide

Page 1: Introduction to Deep Learning

Deep Learning is a subset of Machine Learning that uses Artificial Neural Networks with multiple hidden layers to learn complex patterns from data. It is inspired by the structure and function of the human brain. Deep Learning powers applications such as image recognition, speech recognition, recommendation systems, self-driving cars, natural language processing, and generative AI. The availability of large datasets, powerful GPUs, and advanced algorithms has led to significant breakthroughs in the field.

Page 2: Neural Networks Fundamentals

Artificial Neural Networks (ANNs) consist of input layers, hidden layers, and output layers. Each neuron receives inputs, applies weights and biases, and passes the result through an activation function such as ReLU, Sigmoid, or Tanh. During training, the network adjusts weights using backpropagation and gradient descent to minimize prediction errors. Concepts such as epochs, batch size, learning rate, and loss functions are fundamental to training neural networks effectively.

Page 3: Popular Deep Learning Architectures

Convolutional Neural Networks (CNNs) are primarily used for computer vision tasks such as image classification and object detection. Recurrent Neural Networks (RNNs), LSTMs, and GRUs are designed for sequential data such as text and time-series analysis. Transformers have revolutionized NLP by using self-attention mechanisms, enabling models like GPT, BERT, and modern Large Language Models. Autoencoders and GANs are widely used in generative AI.

Page 4: Training, Evaluation, and Challenges

Training deep learning models requires large amounts of data and computational power. Common optimization algorithms include SGD, Adam, and RMSProp. Evaluation metrics vary by task, including Accuracy, Precision, Recall, F1-Score, BLEU Score, and Perplexity. Challenges include overfitting, vanishing gradients, exploding gradients, high training costs, model interpretability, and bias in datasets. Techniques such as dropout, batch normalization, and regularization help improve performance.

Page 5: Modern Applications and Future Trends

Deep Learning is the foundation of modern AI systems including Large Language Models, Generative AI, Agentic AI, Autonomous Vehicles, Medical Imaging, Fraud Detection, and Recommendation Engines. Emerging trends include multimodal AI, Retrieval-Augmented Generation (RAG), AI agents, efficient model fine-tuning, and edge AI deployment. As organizations increasingly adopt AI solutions, deep learning remains one of the most valuable skills for AI engineers, researchers, and data scientists.